

12 (a)	gain / loss/dB = $10 \lg(P_1/P_2)$	C1	
	$190 = 10 \lg(18 \times 10^3 / P_2)$		
	or $-190 = 10 \lg P_2 / 18 \times 10^3$	C1	
	power = $1.8 \times 10^{-15} \text{ W}$	A1	[3]

(b) (i)	11 GHz / 12 GHz	B1	[1]
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(ii)	e.g. so that input signal to satellite will not be 'swamped' to avoid interference of uplink with / by downlink	B1	[1]
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